

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W010
Well Name	SYN-Assam-Duliajan-010
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.394437 N
Longitude	95.238471 E
Well Status	Abandoned
Well Role	Historical (offset reference well)
Spud Date	26-Feb-2017
Drilling Completed	08-Mar-2017
Report / Completion Date	15-Mar-2017
Total Depth (Driller)	3,260 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E2000-01
Drilling Days	11 days
Recorded Events	7
Recorded NPT	76.8 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W010
Well Name: SYN-Assam-Duliajan-010
Field: SYN-Assam-Duliajan
Latitude: 27.394437
Longitude: 95.238471
Total Depth: 3,260 m
Primary Formation: Sandstone
Well Status: Abandoned
Spud Date: 2017-02-26
Event Count: 7
Total NPT: 76.8 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W010 (SYN-Assam-Duliajan-010) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,260 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,260 m, completed in 11 rotary drilling days between 26-Feb-2017 and 08-Mar-2017. Average overall penetration rate for the well was 12.3 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–708 m), Shale (708–1,859 m), Sandstone (1,859–2,895 m), Limestone (2,895–3,260 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 7 reportable events were recorded, comprising 1 × Casing Issue, 1 × Stuck Pipe, 1 × Wellbore Instability, 4 × Mud Loss. Aggregate non-productive time was 76.8 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 710 m, 9-5/8" at 1,860 m, 7" at 3,260 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Abandoned**. The well was plugged and abandoned after evaluation; cement plugs were set across the open hole and shoe intervals.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W005	SYN-Assam-Duliajan-005	4.86	3,560	Sandstone	5
W002	SYN-Assam-Duliajan-002	5.00	3,450	Sandstone	7
W007	SYN-Assam-Duliajan-007	6.93	3,400	Sandstone	3
W001	SYN-Assam-Duliajan-001	9.58	3,470	Sandstone	4
W008	SYN-Assam-Duliajan-008	10.75	3,690	Sandstone	8
W004	SYN-Assam-Duliajan-004	11.15	3,590	Sandstone	2

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	708	708	Casing Issue recorded (1 event)
2	Shale	708	1,859	1,151	Stuck Pipe, Wellbore Instability, Mud Loss recorded (3 events)
3	Sandstone	1,859	2,895	1,036	Mud Loss recorded (2 events)
4	Limestone	2,895	3,260	365	Mud Loss recorded (1 event)

Formation: Claystone Top: 0 m Bottom: 708 m
Formation: Shale Top: 708 m Bottom: 1859 m
Formation: Sandstone Top: 1859 m Bottom: 2895 m
Formation: Limestone Top: 2895 m Bottom: 3260 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–708	9.58	9.2–10.0	31.1	144	8.2
Shale	708–1,859	10.95	10.3–11.6	17.0	124	16.9
Sandstone	1,859–2,895	12.21	11.5–12.9	17.1	113	24.0
Limestone	2,895–3,260	13.58	13.4–13.8	8.6	91	33.3

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 13.6 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 120 rpm range on average with weight on bit averaging 19.3 klbf. Penetration rate declined with depth from 31.1 m/hr in the upper hole to 8.6 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0058	594	Claystone	Casing Issue	Low	6.6
E0063	1,155	Shale	Stuck Pipe	Medium	13.0
E0062	1,554	Shale	Wellbore Instability	High	20.8
E0061	1,695	Shale	Mud Loss	Medium	7.3
E0060	2,305	Sandstone	Mud Loss	Medium	6.5
E0057	2,628	Sandstone	Mud Loss	High	15.1
E0059	2,952	Limestone	Mud Loss	Medium	7.5

Well ID: W010		Event Type: Casing Issue		Depth: 594 m		Formation: Claystone		Severity: Low		Duration: 6.6 hours
Well ID: W010		Event Type: Stuck Pipe		Depth: 1155 m		Formation: Shale		Severity: Medium		Duration: 13.0 hours
Well ID: W010		Event Type: Wellbore Instability		Depth: 1554 m		Formation: Shale		Severity: High		Duration: 20.8 hours
Well ID: W010		Event Type: Mud Loss		Depth: 1695 m		Formation: Shale		Severity: Medium		Duration: 7.3 hours
Well ID: W010		Event Type: Mud Loss		Depth: 2305 m		Formation: Sandstone		Severity: Medium		Duration: 6.5 hours
Well ID: W010		Event Type: Mud Loss		Depth: 2628 m		Formation: Sandstone		Severity: High		Duration: 15.1 hours
Well ID: W010		Event Type: Mud Loss		Depth: 2952 m		Formation: Limestone		Severity: Medium		Duration: 7.5 hours

7.1 EVENT DETAIL (1–3 OF 7)

EVENT E0058 — CASING ISSUE AT 594 m (CLAYSTONE)

Well ID: W010
Event ID: E0058
Event Type: Casing Issue
Depth: 594 m
Formation: Claystone
Severity: Low
Duration: 6.6 hours
Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Casing integrity concern noted opposite Claystone at 594 m; pressure test anomaly investigated before proceeding. Parameters recorded around the event depth: mud weight 9.85 ppg, ROP 28.6 m/hr, 145 rpm, WOB 10.7 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

EVENT E0063 — STUCK PIPE AT 1,155 m (SHALE)

Well ID: W010
Event ID: E0063
Event Type: Stuck Pipe
Depth: 1155 m
Formation: Shale
Severity: Medium
Duration: 13.0 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Differential/mechanical sticking of the string at 1155 m in Shale; hole cleaning improved and pipe recovered. Parameters recorded around the event depth: mud weight 10.79 ppg, ROP 16.8 m/hr, 124 rpm, WOB 16.5 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0062 — WELLBORE INSTABILITY AT 1,554 m (SHALE)

Well ID: W010
Event ID: E0062
Event Type: Wellbore Instability
Depth: 1554 m
Formation: Shale
Severity: High
Duration: 20.8 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Borehole collapse tendency in Shale at 1554 m; repeated back-reaming needed to maintain hole condition. Parameters recorded around the event depth: mud weight 11.30 ppg, ROP 17.0 m/hr, 123 rpm, WOB 18.5 klbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

7.2 EVENT DETAIL (4–6 OF 7)

EVENT E0061 — MUD LOSS AT 1,695 m (SHALE)

Well ID: W010
 Event ID: E0061
 Event Type: Mud Loss
 Depth: 1695 m
 Formation: Shale
 Severity: Medium
 Duration: 7.3 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Seepage-to-partial mud losses observed while drilling Shale at 1695 m; mud weight trimmed and losses monitored. Parameters recorded around the event depth: mud weight 11.42 ppg, ROP 16.5 m/hr, 116 rpm, WOB 19.8 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0060 — MUD LOSS AT 2,305 m (SANDSTONE)

Well ID: W010
 Event ID: E0060
 Event Type: Mud Loss
 Depth: 2305 m
 Formation: Sandstone
 Severity: Medium
 Duration: 6.5 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Partial loss of drilling fluid recorded at 2305 m in the Sandstone interval; returns reduced and LCM pumped to regain circulation. Parameters recorded around the event depth: mud weight 12.09 ppg, ROP 19.8 m/hr, 111 rpm, WOB 22.6 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0057 — MUD LOSS AT 2,628 m (SANDSTONE)

Well ID: W010
 Event ID: E0057
 Event Type: Mud Loss
 Depth: 2628 m
 Formation: Sandstone
 Severity: High
 Duration: 15.1 hours
 Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Loss of circulation into porous Sandstone at 2628 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 12.62 ppg, ROP 17.0 m/hr, 105 rpm, WOB 26.9 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

7.3 EVENT DETAIL (7–7 OF 7)

EVENT E0059 — MUD LOSS AT 2,952 m (LIMESTONE)
Well ID: W010 Event ID: E0059 Event Type: Mud Loss Depth: 2952 m Formation: Limestone Severity: Medium Duration: 7.5 hours Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned. Description. Loss of circulation into porous Limestone at 2952 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 13.47 ppg, ROP 8.5 m/hr, 96 rpm, WOB 32.4 klb. Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval. Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	710	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	1,860	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 590 m
7"	N-80, 29 lb/ft	3,260	8-1/2"	Tail 15.8 ppg, TOC 1,710 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0058** — Casing Issue at 594 m in Claystone (Low, 6.6 hr). Casing integrity concern noted opposite Claystone at 594 m; pressure test anomaly investigated before proceeding.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,260 m
Primary Formation	Sandstone
Major Events	2 of 7 events rated High or Critical
Total NPT	76.8 hours
Completion Status	Abandoned
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Casing Issue** at 594 m (Claystone) — 6.6 hr cumulative.
- **Stuck Pipe** at 1,155 m (Shale) — 13.0 hr cumulative.
- **Wellbore Instability** at 1,554 m (Shale) — 20.8 hr cumulative.
- **Mud Loss** at 1,695 m, 2,305 m, 2,628 m, 2,952 m (Shale, Sandstone, Limestone) — 36.4 hr cumulative.

9.2 Key Historical Observations

- Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.
- Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.
- Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Allow a dedicated conditioning trip before running casing across 594–594 m and limit running speed through the tight interval.
- Improve hole cleaning and minimise stationary time between 1,155 and 1,155 m; plan a wiper trip before extended surveys in this interval.
- Raise inhibition before entering the Shale section and limit open-hole exposure time; treat rising cavings volume as an early warning.
- Approach the Shale/Sandstone/Limestone interval (1,695–2,952 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.

10. DOCUMENT CONTROL

Document Number	WCR-W010-R0
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Issue Date	15-Mar-2017
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.